

Kjartan Einarsson

Senior Full Stack Software Engineer

- **Office:** Victoria, BC, Canada
 - **Email:** kjartanreinarsson@gmail.com
 - **Home:** <https://www.forritun.dev/>
 - **GitHub:** <https://github.com/KjartanE>
 - **LinkedIn:** <https://www.linkedin.com/in/kjartane>
-

PROFILE

Senior Full Stack Engineer and DevOps specialist who ships digital services for the BC Government and enterprise. His stack spans Vue 3, Nuxt, TypeScript, Vite, and Node.js on the front end, and Python, Flask, PostgreSQL, and RESTful APIs following OpenAPI on the back end. He is hands-on with OpenShift, GitOps/IaC, containerization, and CI/CD pipelines including GitHub Actions. He advises, coaches, and mentors cross-functional teams through pair programming and code reviews, works with stakeholders to create user stories and backlog items, and collaborates with designers and UX to meet user needs. He communicates technical concepts to non-technical audiences and brings a delivery-focused mindset to Scrum and Agile environments.

SKILLS

Core Languages

- JavaScript / TypeScript
- Python
- Java
- C / C++

Front-End Development

- Vue.js (Vue 3)
- Nuxt.js
- Vite
- React
- React-Native
- Next.js
- Angular
- Tailwind CSS
- PostCSS / CSS Modules
- Material-UI
- Shad-CN
- HTML/CSS

Back-End Development

- Node.js
- Python
- Flask
- Django
- Express
- Fast-API
- Spring Boot
- RESTful APIs / OpenAPI 3.x

Database Management

- PostgreSQL
- MySQL

- MongoDB
- DynamoDB
- Redis
- RabbitMQ

DevOps and Deployment

- OpenShift 4.x
- Docker
- Kubernetes
- GitOps / IaC
- Helm
- GitHub Actions
- CI/CD Pipelines
- Azure Kubernetes Cluster

Version Control and Testing

- Git
- Mocha
- Jest
- Cypress
- Selenium
- JMeter
- Automated Testing

Leadership and Mentoring

- Team Mentoring and Development
- Code Review and PR Processes
- Agile/Scrum Methodologies
- Technical Best Practices Coaching
- Cross-functional Team Collaboration

System Administration and Network Management

- Network Management
- Raspberry Pi Development
- Arch Linux
- Debian/Ubuntu

Delivery Methodologies

- Agile Methods
- Scrum
- Hybrid Agile
- Waterfall
- Predictive

PROFESSIONAL EXPERIENCE

Quartech Systems Ltd — September 2020 – Present (November 2025)

BC Government, Digital Trust - TDAC — Oct 2025 – Present

BCGOV – BC Wallet Mobile (BCSC)

Senior Full Stack Developer — Oct 2025 – Present

Working with the CDT (Credential and Digital Trust) team as part of the Integrated Identity Services (IIS) initiative, developing the BCSC (BC Services Card) app and integrating BC Wallet into the BC Services Card experience. Delivering mobile UI, security flows, and backend integration for Android and iOS so citizens can manage digital credentials and access government services in one app.

- Designed and implemented **ScreenWrapper**, a common Bifold component for consistent screen sizing/layout; integrated it across every screen in bc-wallet for unified safe area and scroll behavior
- Implemented BCSC account nickname (create, edit, selection) and persistence to the IAS backend
- Built verification and onboarding flows: setup steps, resume verification, verification approval notifications, and device invalidation / forced de-registration
- Added security method selector with PIN create/update and device permission handling; fixed permission race conditions and Android notification prompts
- Introduced global error handling and consistent alert styling; improved safe area, scroll behavior, WebView scaling, and help/verify-in-person links
- Added i18n support for all BCSC v4 screens
- Improved CI with GitHub Actions for ledgers on PR and SonarCloud; added developer patches for iOS and Android local setup

Technologies: React Native, TypeScript, Bifold (wallet core), Credo-TS (verifiable credentials), Jest, ESLint, Prettier, Yarn workspaces, GitHub Actions, SonarCloud, Swift (iOS), Kotlin (Android)

Internal product — Digital Trust Services (QDTS) — Oct 2024 – Oct 2025

Senior Full Stack Developer

Internal SaaS development for Quartech: Digital Trust Services develop and implement blockchain-based solutions that enable organizations to establish self-sovereign identities and issue, hold, and manage digitally verifiable credentials. These services enhance security, streamline operations, and support government and business entities' digital transformation efforts. Kjartan's responsibilities include:

- Led development of enterprise-scale web applications using React, Node.js, TypeScript, and Kubernetes with a focus on scalability, security, and performance
- Designed and implemented Helm charts and GitHub Actions workflows for five applications in Quartech's Azure Kubernetes Cluster
- Organized the repository by restructuring front-end and back-end components
- Created Docker files for development and production environments for all applications
- Streamlined production deployment by optimizing npm and Yarn packages
- Migrated the API database from AWS DynamoDB to a self-hosted Bitnami chart MongoDB image connected through a Helm chart dependency
- Developed a full Issuer Demo repository, specifically the admin control panel for tenant invitations and user credential management with Keycloak OpenID Connect (OIDC)
- Implemented Keycloak tags to restrict Admin access to Issuer Pages only
- Updated front-end applications to integrate Keycloak OIDC context
- Automated CI and CD pipelines for application deployment and maintenance, notably three repositories and five containers with full CI and CD support

Technologies: Helm, OpenShift, Kubernetes, Azure Kubernetes Cluster, TypeScript, Node.js, Docker, GitHub Actions

Ministry of Water, Land and Resource Stewardship — Dec 2021 – Sep 2024

Northeastern Restoration Tracker (NERT) — May 2024 – Sep 2024

Senior Full Stack Developer/QA Specialist

NERT is a Geographic Information System (GIS)-based web application developed to support restoration efforts in BC's Northeast Natural Resource region. NERT has been expanded to cover all regions of BC, addressing ecosystems that include forests, wetlands, and coastal areas. The application provides interactive map features for visualizing projects, allowing stakeholders to understand spatial distribution and comprehensively track restoration efforts. Kjartan's responsibilities include:

- Provided pivotal support to developing and deploying the NERT application to OpenShift-native environments (DEV/TEST/PROD) using GitOps/IaC practices
- Implemented test automation frameworks (unit, integration, end-to-end) using Mocha, Jest, and Cypress
- Worked effectively with multidisciplinary teams, including Product Owners, Scrum Masters, and Compliance specialists, to create user stories, refine backlog items, and deliver shared outcomes
- Led development of enterprise-scale web applications using React, Node.js, TypeScript, and OpenShift 4.x with focus on scalability, security, and performance
- Redesigned RESTful API systems following OpenAPI conventions, with layered architecture for endpoints, services, and repositories
- Wrote initial database migrations at the DEV, TEST, and PROD stages of development
- Modified Helm charts, YAML manifests, and ENV variables for OpenShift-native deployments
- Integrated authentication and verification throughout the project (OAuth)
- Designed and optimized PostgreSQL for performance, indexing, query optimization, schema design, and migration strategies

- Contributed to open-source and inner-source codebases, promoting reusable patterns, shared libraries, and collaborative coding standards

Technologies: React and Material-UI, Node.js, TypeScript, Git, PostgreSQL, Docker, OpenShift, Mocha, Jest

Species Inventory Management System (SIMS)/BioHub (BIOHUB) — Dec 2021 – Apr 2024

Senior Full Stack Developer

SIMS (also referred to as BioHub) is a digital transformation initiative to centralize and modernize the management of aquatic and terrestrial species and habitat data in BC. This project streamlined data collection, management, and sharing processes to enhance data accessibility and decision-making capabilities for biologists, consultants, and government agencies. The system integrated multiple legacy systems into a unified platform to improve environmental management and policy development. Kjartan's responsibilities included:

- Led development of enterprise-scale web applications using React, Node.js, TypeScript, and OpenShift with focus on scalability, security, and performance
- Redesigned RESTful API systems following OpenAPI conventions, with layered architecture for endpoints, services, and repositories
- Wrote database migrations at the DEV, TEST, and PROD stages of development
- Modified Helm charts, YAML manifests, and ENV variables for OpenShift-native deployments
- Integrated authentication and verification throughout the project (OAuth)
- Implemented test automation frameworks (unit, integration, end-to-end) using Mocha and Jest
- Developed savers and parsers for CSV, XML, and XLS files
- Leveraged token validation to preserve user authentication
- Created a front-end map using Leaflet
- Completed development and migration from TEST-PROD and DEV-TEST

Technologies: React/Material-UI, NodeJS, TypeScript, Git, PostgreSQL, Docker, OpenShift, Mocha, Jest

Habitat Restoration Tracker (HRT) — Jan 2023 – Apr 2023

Senior Full Stack Developer

Kjartan served as a full-stack developer during the foundational development phase of the HRT project. This critical project developed a platform aimed at managing restoration efforts across BC. Kjartan's responsibilities included:

- Assisted in the redesign and integration of the RESTful API system
- Implemented test automation frameworks (unit, integration, end-to-end) for back-end and front-end testing
- Established the initial project setup and development lifecycle
- Integrated scientific data to deliver a transparent and comprehensive overview of land-based restoration activities

Technologies: React and Material-UI, Node.js, TypeScript, Git, PostgreSQL, Docker, OpenShift, Mocha, Jest

Investment Agriculture Foundation (BuyBC) — Jan 2023 – Apr 2023

Full Stack Developer

This project was developed to demonstrate the issuance of verifiable credentials to organizations in OrgBook BC. BuyBC uses the OrgBook DEV API for search functionality, allowing users to issue a BuyBC credential linked to an organization's BC registration ID. Kjartan's responsibilities included:

- Built and iterated on Vue.js front-end features for the BuyBC issuer and verifier applications, applying component-based architecture and TypeScript for type safety
- Implemented a user feedback form and optimized search bar interactions for improved UX
- Collaborated with stakeholders to refine UI requirements and deliver responsive, accessible front-end updates
- Assisted in the migration of the BuyBC application to IAF servers, updating Docker configurations and environment variables for the target environment

Technologies: Vue.js (Vue 3), TypeScript, Docker, Git

SYNQ — Jun 2020 – Dec 2020

Full Stack Developer

SYNQ is a technology company specializing in providing real-time integration and analytics solutions. The SYNQ platform facilitates seamless data synchronization across different sources, enabling businesses to make informed, strategic decisions based on the most current information available. Kjartan's responsibilities included:

- Designed and updated the company's website
- Developed and installed a network check-in software for company assets and devices
- Developed a prototype for a remote access locker system and wiring, soldering, and programming the device
- Provided technology resolution to clients through verbal direction

- Delivered in-person tech support for client devices and assets

Technologies: Python, MySQL, HTML, CSS, Network Management, Raspberry Pi Development, Arch Linux

EDUCATION

Bachelor of Software Engineering

University of Victoria | September 2017 – April 2024

CERTIFICATIONS

- AWS Cloud Practitioner — Amazon Web Services | April 2024
 - OpenShift 101/201 — BC Government | April 2024
 - Certified Practitioner in Agile Development Practices
-

AWARDS

- Victoria Lions Club Directors Scholarship
 - University of Victoria Entrance Scholarship
-

PERSONAL PROJECTS

Art, Genuary & Pen Plotter

- Participating in Genuary with a daily generative-art framework (canvas-sketch, P5.js, Three.js), exploring L-systems, reaction-diffusion, tilings, and spirographs.
- Designed and built a custom pen plotter end-to-end: mechanical design, motion control, and firmware, with repeated iterations on frame, belts, and pen lift for better accuracy and line quality.
- Wired the pipeline from code to paper: Genuary sketches export G-code tuned to the machine, so generative pieces run from browser to physical plot.

Music Production and Hardware Synth Design

- Currently designing and building hardware synthesizers, exploring innovative approaches to sound synthesis and music creation.